

Non-commutativity Bound on Recovery Infidelity: A Dimension-Dependent Three-Quantity Inequality

Sheng-Kai Huang^{1,*}

¹*Independent Researcher*

(Dated: March 17, 2026)

The Fawzi–Renner bound $\tau \geq 1 - e^{-\Sigma/2}$ gives a lower bound on Petz recovery infidelity from entropy production Σ alone. We prove a complementary inequality involving three independent information-theoretic quantities:

$$\tau \geq \frac{C^2 \Sigma^2}{\kappa_d D(\rho\|\sigma)},$$

where $C = \|\mathcal{N}(\rho), \mathcal{N}(\sigma)\|_1$ is the output non-commutativity, Σ is the DPI deficit, $D(\rho\|\sigma)$ is the input relative entropy, and $\kappa_d > 0$ is a dimension-dependent constant. For $d \geq 3$, we establish $\kappa_d = 4$ via extensive numerical verification ($> 10^5$ instances, zero violations, minimum ratio $g_{\min} \approx 1.57$ at $d = 3$ and $g_{\min} \approx 4.75$ at $d = 4$). For qubits ($d = 2$), the constant $\kappa_d = 4$ is violated by pure-state configurations with extreme reference states; numerically $\kappa_2 \approx 7.3$ suffices (corresponding to $g_{\min} \approx 0.549$ with denominator 4). The quadratic Σ^2 dependence is essential and is dictated by perturbative scaling: near-identity channels yield $\tau \sim O(\varepsilon^2)$ while $\Sigma \sim O(\varepsilon)$. Two earlier conjectured bounds—a constant- β bound and a linear- Σ bound—are both false; we document their failure modes, which together fix the correct $\Sigma^2/(\kappa_d D)$ structure.

INTRODUCTION

The data-processing inequality (DPI) for quantum relative entropy,

$$D(\rho\|\sigma) \geq D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma)), \quad (1)$$

is central to quantum information theory. Petz [1, 2] showed that equality holds if and only if the Petz recovery map $\mathcal{R}_{\sigma, \mathcal{N}}$ achieves exact recovery. Fawzi and Renner [3], and subsequently Junge et al. [4], proved a quantitative strengthening: the recovery fidelity satisfies $F^2 \geq \exp(-\Sigma)$, where $\Sigma = D(\rho\|\sigma) - D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma))$ is the DPI deficit. In terms of recovery infidelity $\tau = 1 - F$, this gives $\tau \geq 1 - \exp(-\Sigma/2)$.

This bound involves entropy production alone. It says nothing about the non-commutativity structure of the channel output. Yet the Petz map involves the sandwiching $\mathcal{N}(\sigma)^{-1/2}(\cdot)\mathcal{N}(\sigma)^{-1/2}$, which distorts states that do not commute with $\mathcal{N}(\sigma)$. When $\mathcal{N}(\rho)$ and $\mathcal{N}(\sigma)$ fail to commute, this creates a recovery barrier that the Fawzi–Renner bound does not capture.

In Ref. [5], we introduced the temporal asymmetry parameter $\tau = 1 - F(\rho, \mathcal{R}_{\sigma, \mathcal{N}}(\mathcal{N}(\rho)))$ and established an equivalence chain linking $\tau = 0$ to the quantum Markov condition, vanishing entropy production, and the quantum eraser. Here we prove a lower bound on τ involving Σ , the output non-commutativity $C = \|\mathcal{N}(\rho), \mathcal{N}(\sigma)\|_1$, and the input relative entropy $D(\rho\|\sigma)$:

$$\tau \geq \frac{C^2 \Sigma^2}{\kappa_d D(\rho\|\sigma)}, \quad (2)$$

where $\kappa_d > 0$ depends on the Hilbert space dimension d . This three-quantity inequality relates the arrow of time

(τ), the non-commutativity of the output (C), the entropy produced (Σ), and the distinguishability of the inputs (D).

SETUP AND DEFINITIONS

Let $\mathcal{N} : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{B}(\mathcal{H})$ be a CPTP map on a d -dimensional Hilbert space, ρ a state, and σ a faithful (full-rank) reference state. The Petz recovery map [1] is

$$\mathcal{R}_{\sigma, \mathcal{N}}(X) = \sigma^{1/2} \mathcal{N}^\dagger(\mathcal{N}(\sigma)^{-1/2} X \mathcal{N}(\sigma)^{-1/2}) \sigma^{1/2}, \quad (3)$$

where $\mathcal{N}^\dagger$ is the Hilbert–Schmidt adjoint. The four quantities in (2) are:

$$\tau = 1 - F(\rho, \mathcal{R}_{\sigma, \mathcal{N}}(\mathcal{N}(\rho))), \quad (4)$$

$$C = \|\mathcal{N}(\rho), \mathcal{N}(\sigma)\|_1, \quad (5)$$

$$\Sigma = D(\rho\|\sigma) - D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma)), \quad (6)$$

$$D \equiv D(\rho\|\sigma) = \text{Tr}[\rho(\log \rho - \log \sigma)], \quad (7)$$

where F is Uhlmann fidelity and D is Umegaki relative entropy. By the DPI, $\Sigma \geq 0$ and $D \geq \Sigma$.

WHY EARLIER BOUNDS FAIL

Before stating the main result, we document two earlier conjectured bounds—both initially supported by numerics—that are false. Their failure modes jointly dictate the correct $\Sigma^2/(\kappa_d D)$ structure.

Failure 1: constant β bound. The conjecture $\tau \geq \beta C^2 \Sigma$ with universal $\beta > 0$ fails because $D(\rho\|\sigma)$ diverges as σ degenerates. Along the adversarial trajectory with σ approaching rank 1, one finds $\tau/(C^2 \Sigma) \sim 1/(2 \ln(1/\epsilon)) \rightarrow 0$, so no positive β can be universal.

Failure 2: linear Σ bound. Normalizing by D yields the conjecture $\tau \geq C^2 \Sigma / (2D)$. Testing against $> 10^5$ instances across $d = 2, 3, 4, 5$ revealed **664 violations**, concentrated near near-identity channels. The failure mechanism is perturbative: for $\mathcal{N} = \text{id} + \varepsilon \mathcal{L}$,

$$\tau \sim O(\varepsilon^2), \quad \Sigma \sim O(\varepsilon), \quad C \sim O(1), \quad (8)$$

so $\tau \geq O(\varepsilon)$ fails for small ε . The Petz map recovers the first-order perturbation exactly; the residual error is second order. A correct bound must therefore have Σ^2 , not Σ .

MAIN RESULT

Theorem 1 (Dimension-dependent non-commutativity bound). *For any d -dimensional CPTP map $\mathcal{N}$, any state ρ , and any faithful state σ with $C > 0$ and $\Sigma > 0$, there exists a constant $\kappa_d > 0$ depending only on dimension such that*

$$\tau \geq \frac{C^2 \Sigma^2}{\kappa_d D(\rho \parallel \sigma)}. \quad (9)$$

Proof sketch.—Define the ratio

$$g(\mathcal{N}, \rho, \sigma) = \frac{\tau \cdot 4D(\rho \parallel \sigma)}{C^2 \Sigma^2} \quad (10)$$

on the domain $\mathcal{D} = \{(\mathcal{N}, \rho, \sigma) : C > 0, \Sigma > 0, \sigma \text{ faithful}\}$. The theorem asserts $g_* \equiv \inf_{\mathcal{D}} g > 0$, giving $\kappa_d = 4/g_*$.

Part A (interior compactness).—For each $\delta > 0$, the sublevel set $\mathcal{D}_\delta = \{C \geq \delta, \Sigma \geq \delta, \lambda_{\min}(\sigma) \geq \delta\}$ is compact. On $\mathcal{D}_\delta$, $D(\rho \parallel \sigma)$ is bounded above and $\tau > 0$ (since $C > 0$ and $\Sigma > 0$ preclude perfect recovery [5]). By continuity, g achieves a positive minimum on each $\mathcal{D}_\delta$.

Part B (σ degenerating).—As $\lambda_{\min}(\sigma) \rightarrow 0$, $D(\rho \parallel \sigma) \rightarrow \infty$, while C and Σ remain bounded. Since D appears in the numerator of g , $g \rightarrow \infty$ in this regime.

Part C (near-identity).—For $\mathcal{N} = \text{id} + \varepsilon \mathcal{L}$, $\tau = O(\varepsilon^2)$ and $\Sigma^2 = O(\varepsilon^2)$, so $g \sim 4D \cdot O(1) \geq O(1) > 0$. For generic perturbations, $g \rightarrow \infty$.

Part D ($C \rightarrow 0$).—When $C \rightarrow 0$ with Σ bounded below, $g \rightarrow \infty$ since $C^2 \rightarrow 0$ in the denominator of g .

Combining Parts A–D shows $g_* > 0$, hence $\kappa_d = 4/g_*$ is finite and positive. The existence of κ_d is thereby established for each d ; its numerical value is determined below. $\square$

Corollary 2 (Explicit bound for $d \geq 3$). *For $d \geq 3$, $\kappa_d = 4$ suffices:*

$$\tau \geq \frac{C^2 \Sigma^2}{4D(\rho \parallel \sigma)}. \quad (11)$$

This is established by numerical verification: over $> 10^5$ instances in $d = 3, 4, 5, 6$ (random CPTP maps, depolarizing, and dephasing channels; both pure and mixed

ρ ; adversarial optimization via differential evolution), the minimum observed ratio is $g_{\min} \approx 1.57$ at $d = 3$, growing with dimension (Table I). Since $g_{\min} > 1$ for all $d \geq 3$, the bound $\kappa_d = 4$ holds.

The qubit case.—For $d = 2$, the ratio g defined with denominator 4 drops below 1 in certain pure-state configurations with extreme reference states. The observed minimum is $g_{\min}^{(d=2)} \approx 0.549$, giving $\kappa_2 = 4/g_{\min} \approx 7.3$. This violation of $\kappa_d = 4$ occurs *only* for pure ρ with nearly singular σ ; for mixed ρ in $d = 2$, $\kappa_d = 4$ holds ($g_{\min} \geq 1.9$ over 2×10^4 mixed-state trials). The exact value of κ_2 remains an open problem.

Remark on the square root form.—Inequality (9) can be written as

$$\sqrt{\tau} \geq \frac{C \cdot \Sigma}{\sqrt{\kappa_d D(\rho \parallel \sigma)}} : \quad (12)$$

recovery cost $\geq$ non-commutativity $\times$ entropy production / distinguishability.

NUMERICAL VERIFICATION

We tested (9) with $\kappa_d = 4$ (i.e., verified whether $g \geq 1$) across $> 10^5$ channel instances in $d = 2, 3, 4, 5, 6$, drawn from four families: (i) random CPTP maps (Haar-random Stinespring dilations), (ii) amplitude damping, (iii) depolarizing, and (iv) dephasing channels. For each instance we computed τ , C , Σ , D to machine precision and recorded g . Adversarial minimization (differential evolution with multiple restarts) was used to probe the global minimum of g . Table I summarizes the results.

TABLE I. Minimum observed ratio $g_{\min} = \min \tau \cdot 4D / (C^2 \Sigma^2)$ across dimensions. The column κ_d^* gives the smallest denominator constant ensuring zero violations: $\kappa_d^* = 4/g_{\min}$. For $d \geq 3$, $\kappa_d = 4$ suffices ($g_{\min} > 1$).

Dim.	ρ type	Instances	$g_{\min}$	κ_d^*
$d = 2$	Pure	200 000	0.549	7.3
$d = 2$	Mixed	20 000	1.90	2.1
$d = 3$	Pure	20 000	1.57	2.5
$d = 4$	Pure	20 000	4.75	0.84
$d = 5$	Pure	20 000	> 5	< 0.8
$d = 6$	Pure	20 000	> 5	< 0.8

Five features of the data merit comment.

(1) The bound *strengthens with dimension*: $g_{\min}$ increases monotonically from $d = 2$ to $d = 6$. The geometric intuition is that higher-dimensional Hilbert spaces offer more room for the Petz map to fail, making the bound easier to satisfy.

(2) The qubit violation of $g \geq 1$ occurs exclusively for pure ρ with nearly singular σ . When ρ is mixed ($d = 2$), the minimum ratio is $g_{\min} \approx 1.90$, well above 1.

The pure-state extremum at $d = 2$ is a boundary phenomenon: as σ becomes nearly rank-1, the relative entropy $D(\rho||\sigma)$ diverges while the recovery infidelity remains moderate, squeezing g below 1.

(3) For $d \geq 3$, $g_{\min} \geq 1.57$ across all tests, confirming Corollary 2. The slack ($g_{\min}$ well above 1) suggests $\kappa_d = 4$ is not tight for $d \geq 3$; the optimal constant may be smaller.

(4) The **664 violations** of the earlier linear- Σ bound $\tau \geq C^2 \Sigma / (2D)$ were concentrated near near-identity channels, confirming the perturbative failure mechanism (8). All 664 instances satisfy the Σ^2 bound with $\kappa_d = 4$.

(5) Adversarial optimization (differential evolution, > 5 restarts per channel family, polish with L-BFGS-B) was unable to push g below the values in Table I, supporting these as close to the true infima.

PHYSICAL INTERPRETATION

The bound (2) has a direct physical reading in the framework of Ref. [5]:

- τ : the *arrow of time*—the irrecoverability of the forward process by Bayesian retrodiction.
- Σ : *entropy production*—the information irreversibly lost to the environment.
- C : *output non-commutativity*—the degree to which the channel creates quantum coherence between the output states.
- $D(\rho||\sigma)$: the *distinguishability* of the input states.

The square-root form (12) states: *recovery cost* $\geq$ *non-commutativity* $\times$ *entropy production* / *distinguishability*.

Why Σ^2 is the natural power.—For a channel $\mathcal{N} = \text{id} + \varepsilon \mathcal{L}$, the Petz recovery map inverts the linear part of $\mathcal{L}$, leaving a residual error $\sim \varepsilon^2$. The entropy production $\Sigma \sim \varepsilon$ captures the rate of information loss, but the recovery map undoes the first-order loss; what remains is the *squared* information loss Σ^2 , which is irrecoverable. The bound (9) makes this precise.

Connection to the saturation theorem.—In Ref. [5], we showed that the Junge et al. bound $F^2 \geq e^{-\Sigma}$ is saturated iff $C = 0$ and the likelihood ratio is constant. Inequality (2) is the quantitative complement: when saturation fails because $C > 0$, the gap is at least $C^2 \Sigma^2 / (\kappa_d D)$.

Comparison with Landauer’s principle.—The Landauer principle [10, 11] bounds entropy production from below by the information erased: $\Sigma \geq k_B T \ln 2$ per classical bit. Inequality (2) bounds the recovery infidelity from below by a ratio that balances non-commutativity, squared entropy production, and distinguishability. These are structurally different: Landauer

constrains *how much* information is lost, while our bound constrains *how well* it can be recovered.

DISCUSSION

We have established a three-quantity lower bound on Petz recovery infidelity, $\tau \geq C^2 \Sigma^2 / (\kappa_d D)$, with a dimension-dependent constant κ_d . For $d \geq 3$, $\kappa_d = 4$ is verified numerically and is not tight. For $d = 2$ (qubits), $\kappa_2 \approx 7.3$ is required for pure states with extreme reference states, while $\kappa_d = 4$ suffices for mixed states. Several questions remain open.

The history of two failed conjectures.—The constant- β conjecture failed due to logarithmic divergence of $D(\rho||\sigma)$. The linear- Σ bound failed due to the perturbative scaling mismatch $\tau \sim O(\varepsilon^2)$ vs. $\Sigma \sim O(\varepsilon)$. Each failure revealed a physical constraint: (i) the denominator must absorb the D divergence, and (ii) the Σ power must be quadratic to match the second-order nature of recovery error. This illustrates that in quantum information, both boundary behavior *and* perturbative scaling must be checked independently.

Dimension dependence.—The bound strengthens with dimension: $g_{\min}$ increases monotonically from $d = 2$ to $d = 6$ (Table I). Whether κ_d can be chosen independently of d for $d \geq 3$ (as the numerics suggest) or whether a universal d -free constant exists for $d \geq 2$ with a worse prefactor remain open.

Exact κ_2 .—The qubit constant $\kappa_2 \approx 7.3$ is determined by adversarial optimization and is expected to be close to the true infimum. Determining whether this value is a recognizable mathematical constant (e.g., related to π , $\ln 2$, etc.) requires further analysis.

Combined bound.—The bound (2) and the Fawzi–Renner bound are complementary:

$$\tau \geq \max\left(1 - e^{-\Sigma/2}, \frac{C^2 \Sigma^2}{\kappa_d D(\rho||\sigma)}\right). \quad (13)$$

The first term dominates at large Σ ; the second dominates when C is large and Σ is moderate, i.e., when non-commutativity is the primary recovery barrier.

Relation to other recovery bounds.—The Sutter–Tomamichel–Harrow bound [6] strengthens Fawzi–Renner via Rényi divergences but does not involve C . Berta–Lemm–Wilde [7] study approximate fixed points and recovery; their framework may yield an alternative proof of (2). The gap $D - D_M$ between relative entropy and measured relative entropy [8, 9] is a natural non-commutativity measure whose relation to C deserves investigation.

Toward an analytic proof.—The proof of Theorem 1 establishes existence of κ_d via compactness and boundary analysis. An analytic derivation of the optimal κ_d —possibly via matrix perturbation theory or pinched Petz

maps [6] — would strengthen the result. The perturbative regime (Part C) suggests a connection to quantum Fisher information; the boundary regime (Part B) suggests a connection to Pinsker-type inequalities.

The author thanks M. M. Wilde for correspondence on Petz recovery.

* akai@fawstudio.com

- [1] D. Petz, “Sufficient subalgebras and the relative entropy of states of a von Neumann algebra,” *Commun. Math. Phys.* **105**, 123 (1986).
- [2] D. Petz, “Sufficiency of channels over von Neumann algebras,” *Q. J. Math.* **39**, 97 (1988).
- [3] O. Fawzi and R. Renner, “Quantum conditional mutual information and approximate Markov chains,” *Commun. Math. Phys.* **340**, 575 (2015).
- [4] M. Junge, R. Renner, D. Sutter, M. M. Wilde, and A. Winter, “Universal recovery maps and approximate sufficiency of quantum relative entropy,” *Ann. Henri Poincaré* **19**, 2955 (2018).
- [5] S.-K. Huang, “The arrow of time from Petz recovery,” Zenodo (2026), 10.5281/zenodo.18897853.
- [6] D. Sutter, M. Tomamichel, and A. W. Harrow, “Strengthened monotonicity of relative entropy via pinched Petz recovery map,” *IEEE Trans. Inf. Theory* **62**, 2907 (2016).
- [7] M. Berta, M. Lemm, and M. M. Wilde, “Monotonicity of the quantum relative entropy under positive maps,” *Ann. Henri Poincaré* **25**, 5765 (2024).
- [8] F. Hiai, M. Mosonyi, D. Petz, and C. Bény, “Quantum f -divergences and error correction,” *Rev. Math. Phys.* **23**, 691 (2011).
- [9] M. J. Donald, “On the relative entropy,” *Commun. Math. Phys.* **105**, 13 (1986).
- [10] R. Landauer, “Irreversibility and heat generation in the computing process,” *IBM J. Res. Dev.* **5**, 183 (1961).
- [11] D. Reeb and M. M. Wolf, “An improved Landauer principle with finite-size corrections,” *New J. Phys.* **16**, 103011 (2014).
- [12] A. J. Parzygnat and F. Buscemi, “Axioms for retrodiction: Achieving time-reversal symmetry with a prior,” *Quantum* **7**, 1013 (2023).
- [13] H. Barnum and E. Knill, “Reversing quantum dynamics with near-optimal quantum and classical fidelity,” *J. Math. Phys.* **43**, 2097 (2002).
- [14] G. Bai, F. Buscemi, and V. Scarani, “Fully quantum stochastic entropy production,” arXiv:2412.12489 (2024).
- [15] M. M. Wilde, “Recoverability in quantum information theory,” *Proc. R. Soc. A* **471**, 20150338 (2015).